



İbrahim Eren Doruk

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PROFESSIONAL SUMMARY

Mechanical Engineer oriented toward R&D and design, with a continuous focus on advancing expertise in manufacturing technologies. Proficient in CAD/CAM, CNC machining, and quality control; conducted research on Selective Laser Melting (SLM) and Thermal Barrier Coating (TBC) technologies. Following full-time employment, currently providing freelance mechanical design, sheet metal manufacturing support, and FDM prototype production services.

EXPERIENCE

Mechanical Design Engineer · Remote / Türkiye

Oct 2025 – Present

Independent Design & Prototyping Engineer

- Delivered client-specific CAD design services; produced 3D models and production-ready technical drawings in SolidWorks and Fusion 360.
- Produced functional FDM prototypes in PLA, PETG, ABS, and ASA materials; managed the entire process from material selection to print parameters.
- Designed sheet metal parts including bend allowance calculations and flat pattern development for laser cutting and press brake operations.

PSM Plaster Systems · Ümraniye, İstanbul

Oct 2024 – Oct 2025

Production Engineer

- Optimized assembly and machining operations in plaster-mortar machine manufacturing to increase production capacity.
- Tracked inventory and material flow to ensure production continuity.
- Prepared and optimized CAM toolpaths in Siemens NX Manufacturing for CNC milling operations; performed 3D part modeling in Fusion 360 in line with engineering change requests.

ESAYSAN Endüstriyel Metal Ürünleri San. · Odunpazarı, Eskişehir

Jul 2024 – Sep 2024

Mechanical Engineering Intern — R&D, Sheet Metal Manufacturing

- Observed and analyzed chipless manufacturing processes including laser cutting, press brake forming, and welded assemblies.
- Took an active role in R&D and Quality Control departments; participated in project management and documentation for leading defense industry companies.

GPS Makine · Hacılar, Kayseri

Jan 2024 – Feb 2024

Mechanical Engineering Intern — Machining

- Performed 3D part design and manufacturing preparation for CNC milling using CAD/CAM software.
- Took an active role in dimensional measurement and inspection activities within quality control operations.

EDUCATION

Eskişehir Osmangazi University · Eskişehir, Türkiye

Graduated 2024

B.Sc. Mechanical Engineering

- Thesis: Surface topology and mechanical properties of AISI 316L stainless steel produced via Selective Laser Melting (SLM).
- Thesis: Thermal barrier coating technologies, application areas, and contributions to aerospace systems.

Eskişehir Osmangazi University · Eskişehir, Türkiye

2019 – 2020

English Preparatory Program

COURSES & CERTIFICATIONS

- TEI TUSAŞ Engine Industries Inc. Training – Spring 2024: Turbine engine design, thermal analysis, manufacturing and quality control.
- Automotive Summer Camp – July 2024: Technical training on hydrogen-fueled and electric vehicles.
- Lean Manufacturing – Engineering Academy (Udemy)
- Sheet Metal Design – Harun Reşit Altunok (Udemy)

TECHNICAL SKILLS

CAD / CAM: SolidWorks, Autodesk Fusion 360, Siemens NX, Siemens NX Manufacturing

Manufacturing: CNC Machining, Sheet Metal Forming, Laser Cutting, Press Brake, Welded Assembly

Additive Mfg.: FDM 3D Printing (PLA, PETG, ABS, ASA), Rapid Prototyping, SLM — research level

Quality & Metrology: Quality Control, Tolerance Analysis, GD&T, Technical Drawing, Dimensional Inspection

Materials: Material Science, Heat Treatment, Surface Coating Technologies

Languages: Turkish (native) | English (B1 — Intermediate) | German (B1 — Intermediate)

REFERENCES

GPS Makine

Okan Gündüz — Mechanical Engineer, GPS Makine

Eskişehir Osmangazi University

Dr. Mehmet Erol Kara — Research Associate Dr., Eskişehir Osmangazi University

ESAYSAN Endüstriyel Metal Ürünleri San.

Canel Güngör — Team Leader, ESAYSAN

Abdulkadir Durukan — R&D Engineer, ESAYSAN

PSM Plaster Systems

Saadet Doruk — Manager, PSM Plaster Systems